The Syntax of Mandarin *Num-Cl P Num-Cl* Constructions

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Puzzles

➢ **Puzzle 1**: Subject-Targeted (ST) or Object-Targeted (OT)? Both !

(1) [tamen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.
'[They]$_j$ eat [apples]$_i$ [one by one]$_{j/i}$.'

➢ **Puzzle 2**: A Deictic Asymmetry between Subject and Object !

Subject-Targeted: 🍎 👦👦👦👦👦👦👦👦

🤔 Partitive Quantification: Just part of them eat apples.
🤔 Universal Quantification: Every single person of them all eat apples.

Object-Targeted: 🍎🍎🍎🍎🍎🍎🍎🍎🍎🍎 ……

🤔 Partitivity Quantification: Just part of apples is eaten up.
😱 We can never have the reading that all apples are eaten up.

➢ **Puzzle 3**: Binding but not C-Commanding !

The object (🍎) at the lowest does not c-command *yige jie yige*, but we can still get the object-targeted reading through the binding relationship between the two, which strongly suggests that the surface structure is not sufficient to derive such a referential connection and *yige jie yige* is supposed to undergo reconstruction.

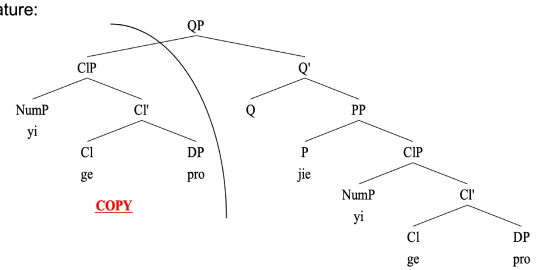
➢ **Puzzle 4**: When we have another quantificational elements (e.g. ModalP or Negation), Object-Targeted Reading Disappears !

(2) a. [tamen]$_j$ bixu/meiyou [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[they-Pl]$_j$ must/Neg [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.
'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/i}$.'
(Modal/Neg > *yige jie yige*: [Subject] √ [Object] √)

b. [tamen]$_j$ [yige jie yige]$_{j/*i}$ de bixu/meiyou chi [pingguo]$_i$.
[they-Pl]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE must/Neg eat [apples]$_i$.
'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/*i}$.'
(*yige jie yige* > Modal/Neg: [Subject] √, [Object] ×)

The Internal Syntax

➢ A syntactic reduplication of ClP with the reduplicative head Q bearing its quantity feature:



- Being a typical numeral classifier language, Mandarin requires a matching relationship between classifiers and nominal targets. Hence, in a transitive sentence with an adverbial NCPNC construction, if the subject and object may share the same classifier, the NCPNC construction is ambiguous between a ST and an OT reading.

😌 **Puzzle 1** ✅

A 4-way Ambiguity

➢ Differentiating SOA & MA:

◆ **Subject-Oriented Adv.** (SOA): as subject-modifier to emphasize the feature of the subject - Universal Quantification: we are sure that everyone of them has eaten.

◆ **Manner Adv.** (MA): as verb-modifier to indicate the pure manner of the event - Distributivity: we don't know if everyone of them all has eaten, what we know is the sequential manner in which they eat.

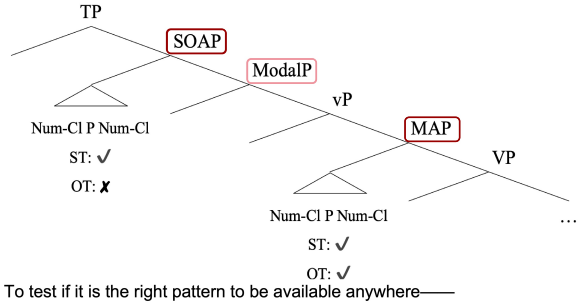
➢ A single NCPNC construction may show a four-way (2*2) ambiguity:

(3) [tamen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.
'[They]$_j$ eat [apples]$_i$ [one by one]$_{j/i}$.'
(i) 'Every single person of them all eat one apple after another.' [SOA-OT]
(ii) 'Every single person of them in turn, all eat apples.' [SOA-ST]
(iii) 'They (possibly together) eat one apple after another.' [MA-OT]
(iv) 'They, one person after another, eat apples in turn.' [MA-ST]

😌 **Puzzle 2** ✅

The External Syntax: Surface Position - A Cartographic View

➢ It is observed that the linear order between an NCPNC construction and another quantificational element, e.g. modal or negation correlates with the availability of the abovementioned four readings:



➢ To test if it is the right pattern to be available anywhere——

◆ Ensuring ST reading with an intransitive verb:

(4) a. tamen bixu yige jie yige de likai.
they must one-Cl following one-Cl DE leave
'They must leave in a manner of one by one.'
(Modal > *yige jie yige*, MA √, SOA ×)

b. tamen yige jie yige de bixu likai.
they one-Cl following one-Cl DE must leave
'One by one, they all must leave.'
(*yige jie yige* > Modal , MA × , SOA √)

◆ Ensuring OT reading with a singular subject:

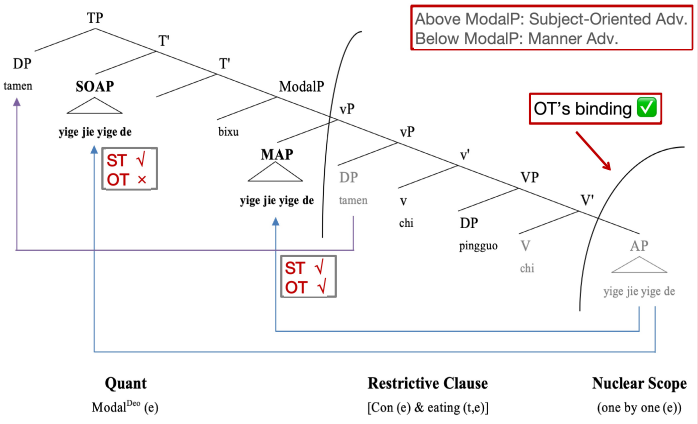
(5) a. wo bixu yipian jie yipian de kan wenxian.
I must one-Cl following one-Cl DE read paper.
'I must read paper in a manner of one by one'
(Modal > *yipian jie yipian*, MA √, SOA ×)

b. * wo yipian jie yipian de bixu kan wenxian.
I one-Cl following one-Cl DE must read paper.
(*yipian jie yipian* > Modal , ungrammatical!)

😌 Matches! But...Puzzle 3, 4: 🤔

The External Syntax: Reconstruction and Epiphenomenon

➢ **Original Place and Reconstruction:** Adopting Larson's (2004, 2014, 2018) proposal, I argue that the NCPNC construction is base-generated in a V-complement position, whereas "*de*" behaves like a functional head ("concordializer") and forces the NCPNC construction to move to its surface position to receive Case through Agreement - a Probe-Goal mechanism. Hence, the OT reading is explained by assuming a binding relationship between *yige jie yige* and the c-commanding object antecedent after reconstruction:



😌 **Puzzle 3** ✅

➢ **Scopal Interaction and Epiphenomenon**: When NCPNC moves to the position under the scope of a quantificational element, both subject-targeted and object-targeted reading are allowed (MA-ST, MA-OT; 6b). When it moves across the quantificational phrase, only SOA-ST interpretationis allowed (6a); the OT reading is ruled out due to the lack of reconstruction which would have violated The Isomorphic Principle (Aoun & Li, 1989), i.e., Chinese bans the inverse scope reading in the sentence with more than one quantifier.

(6) a. *[$_{TP}$ tamen [**yige jie yige de**]$_j$ [$_{ModalP}$ bixu [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]].
(*yige jie yige* > Modal, SOA)

b. [$_{TP}$ tamen [$_{ModalP}$ bixu [$_{vP}$ [**yige jie yige de**]$_j$ [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]]].
(Modal > *yige jie yige*, MA)

😌 **Puzzle 4** ✅

Implications

This study therefore supports the V-complement analysis of manner adverb (Alexiadou, 1997; Stroik, 1990, 1992), as well as Larson's Agreement-based movement account; and the analysis for its quantificational scopal relationship reflects the scope rigidity of quantifiers in Chinese.

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Selected Reference: Aoun, J., & Li, Y. H. A. 1989. Scope and constituency. *Linguistic inquiry*, 20(2), 141-172. Jin, J. (2018). *Partition and quantity: Numeral classifiers, measurement, and partitive constructions in Mandarin Chinese*. Routledge. Kubota, A. 2015. Transforming manner adverbs into subject-oriented adverbs: Evidence from Japanese. *Natural Language & Linguistic Theory*, 33, 1019-1046. Larson, R. K. 2018. AP-de (地) Adverbs in Mandarin. *Studies in Chinese Linguistics*, 39(1), 1-28. Stroik, T. (1990). Adverbs as V-sisters. *Linguistic Inquiry*. 21(4). 654-661. Travis, L. 2001. The syntax of reduplication. In North East Linguistics Society. 31(2), 14. Tsai, W-T. D. (2015). On the topography of Chinese modals. *Beyond functional sequence*. 275-294.